

Lie Algebras Coursework 2

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November 2024

1 Introduction

In this coursework I am going to be finding the roots, coroots, and finding an $\mathfrak{sl}(2)$ triple of the Lie algebra $\mathfrak{sp}(2n)$.

The elements of $\mathfrak{sp}(2n)$ consist of $2n \times 2n$ matrices in $\mathfrak{gl}(n)$ satisfying that $X^t J = -JX$ where $X \in \mathfrak{gl}(n)$ and $J = \begin{pmatrix} 0 & \text{Id}_n \\ -\text{Id}_n & 0 \end{pmatrix}$.

The Lie bracket for this Lie algebra is given by the commutator:

$$[-, -] : \mathfrak{sp}(2n) \longrightarrow \mathfrak{sp}(2n)$$

$$[A, B] \longmapsto AB - BA$$

$\forall A, B \in \mathfrak{sp}(2n)$.

Any element $X \in \mathfrak{sp}(2n)$ can be written in the following block form:

$$X = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

where each A, B, C, D are $n \times n$ matrices in $\mathfrak{gl}(n)$.

To find out the restrictions on A, B, C, D are, let's use our equation that defines $\mathfrak{sp}(2n)$:

$$X^t J = \begin{pmatrix} A^t & C^t \\ B^t & D^t \end{pmatrix} \begin{pmatrix} 0 & \text{Id}_n \\ -\text{Id}_n & 0 \end{pmatrix} = \begin{pmatrix} -C^t & A^t \\ -D^t & B^t \end{pmatrix}$$

$$-JX = \begin{pmatrix} 0 & -\text{Id}_n \\ \text{Id}_n & 0 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} -C & -D \\ A & B \end{pmatrix}$$

Equating the two matrices implies that:

$$A^t + D = 0$$

$$B^t = B$$

$$C^t = C.$$

Now, let's find the dimension of $\mathfrak{sp}(2n)$. We will proceed by finding the size of the basis. If we count the possibilities for each entry in each of the blocks we can see that:

- A can be counted as having no restraints on its basis elements, so the number of possible matrices for A is n^2 .
- B has to be symmetric, so no restraints on the diagonal elements but 'transpose pairs' have to be equal. So there are n possible choices for the diagonal matrices and $\frac{(n-1)^2}{2}$ possible choices for the non-diagonal matrices in the basis. Adding these together gives us $\frac{n(n+1)}{2}$.
- C also has to be symmetric, so similar to B, there are $\frac{n(n+1)}{2}$ possible choices for the basis elements again here.
- D depends entirely on A since it has to be skew-symmetric to A, so in this case, D adds nothing to the basis.

Finally, adding up all the possible choices for each block gives us the dimension, which is:

$$n^2 + \frac{n(n+1)}{2} + \frac{n(n+1)}{2} = n^2 + n(n+1) = 2n^2 + n$$

The Cartan subalgebra has dimension n (from the lecture notes) and since it is maximal, we require $2n^2$ eigenvectors.

Firstly, let's find an n -dimensional set of matrices for the basis of the Cartan subalgebra, then use these to find the eigenvectors.

A good candidate for the basis is

$$H_i = \{E_{i,i} - E_{i+n,i+n} | \forall 1 \leq i \leq n\}.$$

Now, let's apply the adjoint map of H_i to an arbitrary elementary matrix. For all $1 \leq a, b \leq 2n$ we have:

$$\begin{aligned} ad_{H_i}(E_{a,b}) &= [E_{i,i} - E_{i+n,i+n}, E_{a,b}] = [E_{i,i}, E_{a,b}] - [E_{i+n,i+n}, E_{a,b}] \\ &= \delta_{i,a}E_{i,b} - \delta_{i,b}E_{a,i} - \delta_{i+n,a}E_{i+n,b} + \delta_{i+n,b}E_{a,i+n} \end{aligned}$$

Case $i = a$:

$$\longrightarrow E_{i,b} - \delta_{i,b}E_{i,i} + \delta_{i+n,b}E_{i,i+n}$$

Case $i + n = b$:

$$\longrightarrow E_{i,i+n} + E_{i,i+n} = 2E_{i,i+n}$$

So $E_{i,i+n}$ is an eigenvector with eigenvalue 2.

Similarly, swapping a and b we get that $E_{i+n,i}$ is an eigenvector with eigenvalue -2 .

Next, let's apply the adjoint map of H_i to a linear combination of two arbitrary elementary matrices to find more eigenvectors. For all $1 \leq a, b, c, d \leq 2n$, and some $u, v \in \mathbb{C}$, we have:

$$\begin{aligned} ad_{H_i}(uE_{a,b} + vE_{c,d}) &= [E_{i,i}, uE_{a,b}] + [E_{i,i}, vE_{c,d}] - [E_{i+n,i+n}, uE_{a,b}] - [E_{i+n,i+n}, vE_{c,d}] \\ &= u\delta_{i,a}E_{i,b} - u\delta_{i,b}E_{a,i} + v\delta_{i,c}E_{i,d} - v\delta_{i,d}E_{c,i} \\ &\quad - u\delta_{i+n,a}E_{i+n,b} + u\delta_{i+n,b}E_{a,i+n} - v\delta_{i+n,c}E_{i+n,d} + v\delta_{i+n,d}E_{c,i+n} \end{aligned}$$

First set of eigenvectors:

Case $a = i$:

$$\rightarrow uE_{i,b} - u\delta_{i,b}E_{i,i} + v\delta_{i,c}E_{i,d} - v\delta_{i,d}E_{c,i} + u\delta_{i+n,b}E_{i,i+n} - v\delta_{i+n,c}E_{i+n,d} + v\delta_{i+n,d}E_{c,i+n}$$

Case $b = j$, for $1 \leq j \leq n$:

$$\rightarrow uE_{i,j} - u\delta_{i,j}E_{i,i} + v\delta_{i,c}E_{i,d} - v\delta_{i,d}E_{c,i} - v\delta_{i+n,c}E_{i+n,d} + v\delta_{i+n,d}E_{c,i+n}$$

Case $c = j + n$:

$$\rightarrow uE_{i,j} - u\delta_{i,j}E_{i,i} - v\delta_{i,d}E_{j+n,i} - v\delta_{i+n,j+n}E_{i+n,d} + v\delta_{i+n,d}E_{j+n,i+n}$$

Case $d = i + n$:

$$\rightarrow uE_{i,j} - u\delta_{i,j}E_{i,i} - v\delta_{i+n,j+n}E_{i+n,i+n} + vE_{j+n,i+n}$$

Case $i \neq j$:

$$\rightarrow uE_{i,j} + vE_{j+n,i+n}$$

so $uE_{i,j} + vE_{j+n,i+n}$ is an eigenvector for some $u, v \in \mathbb{C}$. We require the components of this eigenvector to be in $\mathfrak{sp}(2n)$, so recall the block matrices representation of an element in $\mathfrak{sp}(2n)$. Because of our restrictions on i and j , the $E_{i,j}$ corresponds to the "A block" and the $E_{j+n,i+n}$ corresponds to the "D block" of the element X. D is the negative transpose of the matrix A, and since we already have the transpose part (i and j are swapped in the indexes) we just require the negative part of the definition for this to be an eigenvector in $\mathfrak{sp}(2n)$. Therefore, let $u = 1, v = -1$ and then we have $E_{i,j} - E_{j+n,i+n}$ is an eigenvector in $\mathfrak{sp}(2n)$, with eigenvalue 1, for $1 \leq i, j \leq n$ with $i \neq j$.

Second set of eigenvectors:

Case $a = i$:

$$\rightarrow uE_{i,b} - u\delta_{i,b}E_{i,i} + v\delta_{i,c}E_{i,d} - v\delta_{i,d}E_{c,i} + u\delta_{i+n,b}E_{i,i+n} - v\delta_{i+n,c}E_{i+n,d} + v\delta_{i+n,d}E_{c,i+n}$$

Case $c = j$, for $1 \leq j \leq n$:

$$\rightarrow uE_{i,b} - u\delta_{i,b}E_{i,i} + v\delta_{i,j}E_{i,d} - v\delta_{i,d}E_{j,i} + u\delta_{i+n,b}E_{i,i+n} + v\delta_{i+n,d}E_{j,i+n}$$

Case $b = j + n$:

$$\rightarrow uE_{i,j+n} + v\delta_{i,j}E_{i,d} - v\delta_{i,d}E_{j,i} + u\delta_{i+n,j+n}E_{i,i+n} + v\delta_{i+n,d}E_{j,i+n}$$

Case $d = i + n$:

$$\rightarrow uE_{i,j+n} + v\delta_{i,j}E_{i,i+n} + u\delta_{i+n,j+n}E_{i,i+n} + vE_{j,i+n}$$

Case $i < j$:

$$\rightarrow uE_{i,j+n} + vE_{j,i+n}$$

Let's proceed as before to find the values u and v . The $E_{i,j+n}$ matrix corresponds to the "B block" and the $E_{j,i+n}$ corresponds to the "B block" also. Since $i < j$, the $E_{i,j+n}$ entry lies above the diagonal in B and the $E_{j,i+n}$ entry lies below the diagonal in B, and they are also "transpose pairs" in B. Moreover, B is symmetric so we require both of these to have the same sign. So let $u = 1$, $v = 1$ and then $E_{i,j+n} + E_{j,i+n}$ is an eigenvector with eigenvalue 1, for all $1 \leq i < j \leq n$.

Third set of eigenvectors:

Case $a = n + i$:

$$\rightarrow -u\delta_{i,b}E_{i+n,i} + v\delta_{i,c}E_{i,d} - v\delta_{i,d}E_{c,i} - uE_{i+n,b} + u\delta_{i+n,b}E_{i+n,i+n} - v\delta_{i+n,c}E_{i+n,d} + v\delta_{i+n,d}E_{c,i+n}$$

Case $b = j$, for $1 \leq j \leq n$:

$$\rightarrow -u\delta_{i,j}E_{i+n,i} + v\delta_{i,c}E_{i,d} - v\delta_{i,d}E_{c,i} - uE_{i+n,j} - v\delta_{i+n,c}E_{i+n,d} + v\delta_{i+n,d}E_{c,i+n}$$

Case $c = j + n$:

$$\rightarrow -u\delta_{i,j}E_{i+n,i} - v\delta_{i,d}E_{j+n,i} - uE_{i+n,j} - v\delta_{i+n,j+n}E_{i+n,d} + v\delta_{i+n,d}E_{j+n,i+n}$$

Case $d = i$:

$$\rightarrow -u\delta_{i,j}E_{i+n,i} - vE_{j+n,i} - uE_{i+n,j} - v\delta_{i+n,j+n}E_{i+n,i}$$

Case $i < j$:

$$\rightarrow -vE_{j+n,i} - uE_{i+n,j}$$

Again as before, the $E_{j+n,i}$ matrix corresponds to a component in the "C block" and the $E_{i+n,j}$ matrix corresponds to a component in the "C block" also, and these are also "transpose pairs" in the matrix C. C is symmetric so

we require u and v to have the same sign, so pick $u = 1, v = 1$ and then $E_{j+n,i} + E_{i+n,j}$ is an eigenvector with eigenvalue -1, for all $1 \leq i < j \leq n$.

So now we have found 5 sets of eigenvectors, but have we found all of them?

For the first 2 sets, i runs between 1 and n , so there are n possible choices for the value of i in each set. This gives us n possible eigenvectors in each set, i.e. $2n$ eigenvectors in both of them.

For $E_{i,j} - E_{j+n,i+n}$ for $1 \leq i, j \leq n$ with $i \neq j$, we can use the permutation formula for our possible combinations of i and j :

$${}^nP_2 = \frac{n!}{(n-2)!} = n(n-1) \frac{(n-2)!}{(n-2)!} = n^2 - n$$

So this set has $n^2 - n$ possible eigenvectors.

The last two sets contain all of the off-diagonal elements of the blocks B and C, these blocks are the same size so these sets have the same number of eigenvectors as each other. Each set contains every "transpose pair" of elements, so we can just count how many elements are above or below the diagonal in B or C. This number is $\frac{n^2-n}{2}$, so in both of the sets we have $n^2 - n$ eigenvectors.

Summing up all the eigenvectors gives us

$$2n + n^2 - n + n^2 - n = 2n^2.$$

Therefore we have found all of the eigenvectors of $\mathfrak{sp}(2n)$ and the list of eigenvectors and eigenvalues are:

- $E_{i,j} - E_{n+j,n+i}$, for $i \neq j$, $1 \leq i, j \leq n$, with eigenvalue 1,
- $E_{i,n+j} + E_{j,n+i}$, for $1 \leq i < j \leq n$, with eigenvalue 1,
- $E_{i,n+i}$, $1 \leq i \leq n$, with eigenvalue 2,
- $E_{n+i,j} + E_{n+j,i}$, $1 \leq i < j \leq n$, with eigenvalue -1, and
- $E_{n+i,i}$, $1 \leq i \leq n$, with eigenvalue -2.

To find the roots associated with each eigenvector, we need to apply ad_{H_i} and ad_{H_j} to each one. It's important to note here that using any other index of H will give us zero because of the Kronecker deltas.

Using the restraints set for each eigenvector and the notation from the lecture notes, we find:

$$E_{i,j} - E_{j+n,i+n}: \alpha(H_i) = 1, \alpha(H_j) = -1, \text{ so } \alpha = e_i - e_j;$$

$$E_{i,j+n} + E_{j,i+n}: \beta(H_i) = 1, \beta(H_j) = 1, \text{ so } \beta = e_i + e_j;$$

$$E_{i+n,j} + E_{j+n,i}: (-\beta)(H_i) = -1, (-\beta)(H_j) = -1, \text{ so } -\beta = -(e_i + e_j);$$

$$E_{i,i+n}: \gamma(H_i) = 2, \gamma(H_j) = 0, \text{ so } \gamma = 2e_i;$$

$$E_{i+n,i}: (-\gamma)(H_i) = -2, (-\gamma)(H_j) = 0, \text{ so } -\gamma = -2e_i;$$

where the e_i 's and e_j 's are standard basis vectors, with a 1 in the i -th slot (or j -th) and zeros everywhere else. These are all the roots of $\mathfrak{sp}(2n)$.

Finding the coroots:

Note that the Killing form of $\mathfrak{sp}(2n)$ is $\kappa(X, Y) = 6\text{tr}_{\mathbb{C}^{2n}}(X.Y)$, for $X, Y \in \mathfrak{sp}(2n)$. Then use this to compute:

$$\begin{aligned}\mathbf{i}(H_i)(H_i) &= \kappa(H_i, H_i) = 6\text{tr}(E_{i,i} + E_{i+n,i+n}) = 12 \\ \mathbf{i}(H_j)(H_j) &= \kappa(H_j, H_j) = 6\text{tr}(E_{j,j} + E_{j+n,j+n}) = 12\end{aligned}$$

$\forall i \neq j :$

$$\begin{aligned}\mathbf{i}(H_i)(H_j) &= \kappa(H_i, H_j) \\ &= 6\text{tr}(E_{i,i}E_{j,j} - E_{i,i}E_{j+n,j+n} - E_{i+n,i+n}E_{j,j} + E_{i+n,i+n}E_{j+n,j+n}) \\ &= 0 \\ &= \mathbf{i}(H_j)(H_i)\end{aligned}$$

Where:

$$\begin{aligned}\mathbf{i}: \mathfrak{g}_0 &\rightarrow \mathfrak{g}_0^* \\ H &\mapsto \kappa(H, -)\end{aligned}$$

Using this, let's write the roots in terms of the $\mathbf{i}$ function as in the lecture notes:

$$\begin{aligned}\alpha &= \frac{\mathbf{i}(H_i - H_j)}{12}, \\ \beta &= \frac{\mathbf{i}(H_i + H_j)}{12}, \\ \gamma &= \frac{\mathbf{i}(2H_i)}{12} = \frac{\mathbf{i}(H_i)}{6}.\end{aligned}$$

Finally, let's compute the norms of each root: ($i \neq j$)

$$||\alpha||^2 = (\alpha, \alpha) = \kappa(\mathfrak{i}^{-1}(\alpha), \mathfrak{i}^{-1}(\alpha)) = \frac{1}{144} \kappa(H_i - H_j, H_i - H_j)$$

$$\begin{aligned} \kappa(H_i - H_j, H_i - H_j) &= 6 \cdot \text{tr}((H_i - H_j)(H_i - H_j)) \\ &= 6 \cdot \text{tr}((H_i^2 - H_i H_j - H_j H_i + H_j^2)) \\ &= 6 \cdot \text{tr}((E_{i,i} + E_{i+n,i+n} + E_{j,j} + E_{j+n,j+n})) \\ &= 6 \cdot 4 \\ &= 24 \end{aligned}$$

$$\begin{aligned} \text{So, } ||\alpha||^2 &= \frac{24}{144} \\ &= \frac{1}{6} \end{aligned}$$

Similarly for β :

$$\begin{aligned} ||\beta||^2 &= \frac{1}{144} \kappa(H_i + H_j, H_i + H_j) \\ &= \frac{6}{144} \cdot \text{tr}((H_i^2 + H_i H_j + H_j H_i + H_j^2)) \\ &= \frac{6}{144} \cdot \text{tr}((E_{i,i} + E_{i+n,i+n} + E_{j,j} + E_{j+n,j+n})) \\ &= \frac{6}{144} \cdot 4 \\ &= \frac{24}{144} \\ &= \frac{1}{6} \end{aligned}$$

And for γ :

$$\begin{aligned} ||\gamma||^2 &= \frac{1}{36} \kappa(H_i, H_i) \\ &= \frac{6}{36} \cdot \text{tr}(H_i^2) \\ &= \frac{1}{6} \cdot \text{tr}(E_{i,i} + E_{i+n,i+n}) \\ &= \frac{1}{6} \cdot 2 \\ &= \frac{1}{3} \end{aligned}$$

Thus, the coroots are:

$$\begin{aligned} H_\alpha &= \frac{2i^{-1}(\alpha)}{\|\alpha\|^2} = 2.6 \cdot \frac{H_i - H_j}{12} = H_i - H_j; \\ H_\beta &= H_i + H_j; \\ H_\gamma &= H_i; \end{aligned}$$

Finding an $\mathfrak{sl}(2)$ -triple:

Recall the roots of $\mathfrak{sp}(2n)$ are:

$$\begin{aligned} \alpha &= e_i - e_j, \\ \beta &= e_i + e_j, \\ \gamma &= 2e_i. \end{aligned}$$

Let's construct an $\mathfrak{sl}(2)$ -triple using the α root: take $E_\alpha = E_{i,j} - E_{j+n,i+n}$. Let's also take $F_\alpha = E_{j,i} - E_{i+n,j+n}$ and check if our choices satisfy the definition from the lecture notes:

$$\kappa(E_\alpha, F_\alpha) = \frac{2}{\|\alpha\|^2} = 2 \cdot 6 = 12$$

$$\begin{aligned} \kappa(E_\alpha, F_\alpha) &= 6 \cdot \text{tr}((E_{i,j} - E_{j+n,i+n})(E_{j,i} - E_{i+n,j+n})) \\ &= 6 \cdot \text{tr}(E_{i,j}E_{j,i} - E_{j+n,i+n}E_{j,i} - E_{i,j}E_{i+n,j+n} + E_{j+n,i+n}E_{i+n,j+n}) \\ &= 6 \cdot \text{tr}(\delta_{j,j}E_{i,i} - \delta_{i+n,j}E_{j+n,i} - \delta_{j,i+n}E_{i,j+n} + \delta_{i+n,i+n}E_{j+n,j+n}) \\ &= 6 \cdot \text{tr}(E_{i,i} + E_{j+n,j+n}) \\ &= 6 \cdot 2 \\ &= 12 \end{aligned}$$

So our choices of E_α and F_α work. Then take the coroot $H_\alpha = H_i - H_j$ as before.

Now, let's check that these elements satisfy the commutation conditions:

Compute $[H_\alpha, E_\alpha]$

$$[H_\alpha, E_\alpha] = [H_i - H_j, E_{i,j} - E_{n+j,n+i}] = [H_i, E_{i,j}] - [H_j, E_{i,j}] - [H_i, E_{n+j,n+i}] + [H_j, E_{n+j,n+i}].$$

Now compute each term separately:

$$[H_i, E_{i,j}] = E_{i,j}, \quad [H_j, E_{i,j}] = -E_{i,j}, \quad [H_i, E_{n+j,n+i}] = -E_{n+j,n+i}, \quad [H_j, E_{n+j,n+i}] = E_{n+j,n+i}.$$

Substitute back into computation:

$$\begin{aligned} [H_\alpha, E_\alpha] &= (E_{i,j} - (-E_{i,j})) - (-E_{n+j,n+i} - E_{n+j,n+i}), \\ [H_\alpha, E_\alpha] &= 2(E_{i,j} - E_{n+j,n+i}) = 2E_\alpha. \end{aligned}$$

Compute $[H_\alpha, F_\alpha]$

$$[H_\alpha, F_\alpha] = [H_i - H_j, E_{j,i} - E_{n+i, n+j}] = [H_i, E_{j,i}] - [H_j, E_{j,i}] - [H_i, E_{n+i, n+j}] + [H_j, E_{n+i, n+j}].$$

Now compute each term separately:

$$[H_i, E_{j,i}] = -E_{j,i}, \quad [H_j, E_{j,i}] = E_{j,i}, \quad [H_i, E_{n+i, n+j}] = E_{n+i, n+j}, \quad [H_j, E_{n+i, n+j}] = -E_{n+i, n+j}.$$

Substitute back into computation:

$$[H_\alpha, F_\alpha] = (-E_{j,i} - E_{j,i}) - (E_{n+i, n+j} - (-E_{n+i, n+j})),$$

$$[H_\alpha, F_\alpha] = -2(E_{j,i} - E_{n+i, n+j}) = -2F_\alpha.$$

Compute $[E_\alpha, F_\alpha]$

$$[E_\alpha, F_\alpha] = [E_{i,j} - E_{n+j, n+i}, E_{j,i} - E_{n+i, n+j}].$$

Expand each term:

$$[E_\alpha, F_\alpha] = [E_{i,j}, E_{j,i}] - [E_{i,j}, E_{n+i, n+j}] - [E_{n+j, n+i}, E_{j,i}] + [E_{n+j, n+i}, E_{n+i, n+j}].$$

Now compute each commutator:

$$[E_{i,j}, E_{j,i}] = E_{i,i} - E_{j,j}, \quad [E_{i,j}, E_{n+i, n+j}] = 0, \quad [E_{n+j, n+i}, E_{j,i}] = 0,$$

$$[E_{n+j, n+i}, E_{n+i, n+j}] = -(E_{n+i, n+i} - E_{n+j, n+j}).$$

Substitute back into computation:

$$[E_\alpha, F_\alpha] = (E_{i,i} - E_{j,j}) - 0 - 0 - (-(E_{n+i, n+i} - E_{n+j, n+j})),$$

$$[E_\alpha, F_\alpha] = (E_{i,i} - E_{n+i, n+i}) - (E_{j,j} - E_{n+j, n+j}) = H_i - H_j = H_\alpha.$$

Therefore,

$$[H_\alpha, E_\alpha] = 2E_\alpha, \quad [H_\alpha, F_\alpha] = -2F_\alpha, \quad [E_\alpha, F_\alpha] = H_\alpha.$$

Thus we can conclude that $\{E_\alpha, H_\alpha, F_\alpha\}$ form an $\mathfrak{sl}(2)$ -triple for $\mathfrak{sp}(2n)$